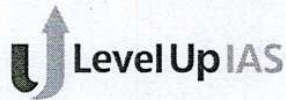


Q.3a (20 marks): Briefly describe Mendel's law of Inheritance. Explain how its misinterpretation made Anthropology a science of discrimination.



(Don't Write anything in this Area)

Q.3) (a) Briefly describe Mendel's law of Inheritance. Explain how its misinterpretation made Anthropology a science of discrimination. (20 marks)

Mendel was the pioneer of genetic study. His study with garden pea in mid 19th c paved the way & laid the foundation of modern genetics.

Mendel's law of inheritance

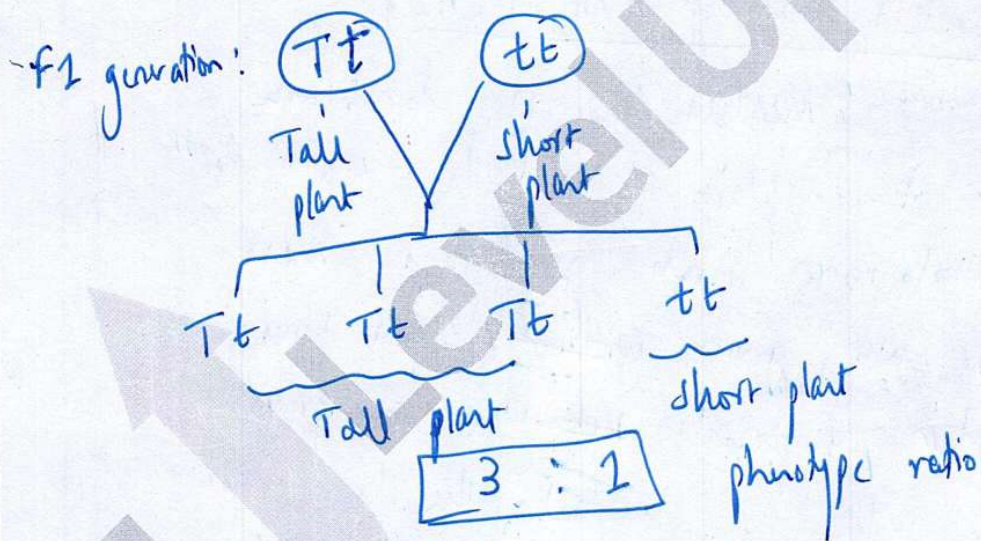
Mendel's law of inheritance tries to explain how the traits are passed on across generations. These laws were given ~~without~~ during times when modern equipments ~~were~~ were not present.

In fact, the laws were once unrecognized & later rediscovered in 1900.

Law of Dominance

It gives us the knowledge of dominant & recessive inheritance. He says that the offspring receives 2 alleles from its

parent and some traits are manifested only when present in both parents i.e. homozygous condition while other traits can express themselves even if present in single parent i.e. heterozygous condition. We called the former as recessive trait & latter as dominant trait.



2) Law of segregation

It says that during gamete formation, the 'factors' of inheritance segregate independently or in equal measure in the ~~trait~~ ~~parent~~ gamete cell can inherit any one of the

allele of their parents

3) Law of independent assortment

It says that the various traits & their factors in humans are inherited independently of each other.

For instance, A plant can be both tall & green or tall & yellow or likewise a short & green ~~or~~ short & yellow i.e. height & color of the plant are independently transmitted.

Misinterpretation of Mendel's law → Anthropology as science of discrimination

1) Mendel's law was grossly misinterpreted on the ~~the~~ complexity of inheritance was not recognised.

2) Every trait was considered unifactorial in nature and single gene was thought to be

- responsible
- 3) Even, social aspects of human like the poverty, illiteracy, etc. was treated as inherited factor
 - 4) Francis Galton even laid the foundation of Eugenics with objective to improve the quality of human population by selecting better traits for reproduction
 - 5) Huxley also gave the concept of social darwinism saying that only fittest should survive and government support for poor, etc. discarded
 - 6) Galton & others even gave simple mathematical method to calculate the probability of inheritance of qualities (i.e.) Pearson's square
 - 7) Discrimination reached peak during world war when Hitler propounded 'Aryans' racial superiority & holocaust of jew was carried out
- However, Modern genetics has now proved beyond doubt the fallacy of race